Medical Insurance Analytics Dashboard

Project Overview

This project focuses on analyzing medical insurance data to uncover insights into healthcare costs, claims, premiums, member demographics, health risks, and regional and network performance.

The project was developed as a Power BI analytics project, with the objective of transforming a large medical insurance dataset into an interactive dashboard that can support data-driven decision-making.

The dataset contains **100,000 member records and 54 variables**, covering information such as member demographics, medical costs, insurance premiums, claims, health conditions, BMI, risk scores, plan types, network tiers, and geographic regions.

Business Problem

Insurance organizations need to understand how healthcare costs and claims vary across different member groups, insurance plans, regions, and risk profiles.

The key questions addressed in this project include:

- How much medical cost is being generated across different regions?
- How do claims compare with insurance premiums?
- Which insurance plans have higher medical costs or claims?
- How is the membership distributed across age groups and genders?
- What proportion of members are classified as high risk?
- How does risk score relate to medical costs?
- How do medical costs and claims vary across network tiers?
- What patterns exist across different member health profiles?

Objectives

The main objectives of the analysis were to:

1. Analyze the overall medical insurance portfolio.
2. Examine medical costs, premiums, and claims.
3. Identify patterns in member health risks.

4. Analyze demographic and geographic differences.
5. Compare insurance plans and network tiers.
6. Build an interactive Power BI dashboard for decision support.

Data Preparation

Before creating the dashboard, the dataset was reviewed to ensure that the analysis matched the actual available fields.

Several analytical fields were created in Power BI where they were not directly available in the source data.

These included:

- **Age Group** – created from the member's age.
- **BMI Category** – created from BMI values.
- **Risk Category** – derived from the existing high-risk indicator.
- **Average Risk Score**
- **High-Risk Member Count**
- **Average Chronic Conditions**
- **Total Claims Count**
- **Claims-to-Premium Ratio**

The analysis deliberately avoided creating unsupported time-series analysis because the dataset did not contain a meaningful date or year field.

Key KPIs

The dashboard tracks several important measures, including:

- **Total Members:** 100,000
- **Total Medical Cost:** approximately 300.95M
- **Total Premium:** approximately 58.23M
- **Total Claims Paid:** approximately 137.79M
- **Average Claim:** approximately 656.51
- **Average Annual Medical Cost:** approximately 3.01K

- **Average Annual Premium:** approximately 582.32
- **High-Risk Members:** 36,781

The claims-to-premium ratio is approximately **236.6%**, providing an important indicator for examining the relationship between claims paid and premium revenue.

Dashboard Structure

The Power BI dashboard consists of five analytical pages:

1. Medical Insurance Executive Overview

Provides a high-level view of the insurance portfolio through key performance indicators and visuals covering:

- Medical cost by region
- Member distribution by plan
- Members by age group
- Member distribution by location type

2. Claims & Cost Analysis

Focuses on the financial performance of the insurance portfolio, including:

- Claims paid by region
- Medical cost versus premium by plan
- Claims paid by age group
- Risk score versus medical cost
- Claims volume by plan

3. Health Risk & Population

Examines member health characteristics and risk patterns, including:

- BMI distribution
- Member risk profile
- Average risk score by region
- Risk score by age group
- Average chronic conditions by region

- Risk profile by gender

4. Member & Plan Analysis

Analyzes the composition of the insured population and differences between plans, including:

- Members by plan type
- Average premium by plan
- Average claim by plan
- Member distribution by gender
- Members by location type and region
- Member age distribution

5. Regional & Network Analysis

Examines geographic and network-level performance through:

- Medical cost by region
- Claims paid by region
- Average risk score by region
- Members by network tier
- Medical cost by network tier
- Claims versus premium by network tier

Tools Used

- **Microsoft Excel** – Data source and initial data inspection
- **Microsoft Power BI** – Data modeling, DAX calculations, visualization, and dashboard development
- **DAX** – Calculated measures and analytical columns
- **GitHub** – Project documentation and portfolio presentation

Key Skills Demonstrated

This project demonstrates practical skills in:

- Data cleaning and validation

- Exploratory data analysis
- Data modeling
- DAX
- KPI development
- Data visualization
- Dashboard design
- Business intelligence
- Healthcare/insurance analytics
- Insight communication
- Data-driven decision-making

Project Outcome

The final dashboard transforms a large insurance dataset into an interactive analytical tool that allows users to explore healthcare costs, claims, premiums, member characteristics, health risks, plans, regions, and network performance.

The project demonstrates how data analytics and visualization can be used to move from raw insurance records to meaningful business insights and support more informed decisions around risk, cost, claims, and member populations.